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PROJECT REPORT

DETECT, TRACK, SECURE

FINNOVATION'25

HACKATHON BY SBI

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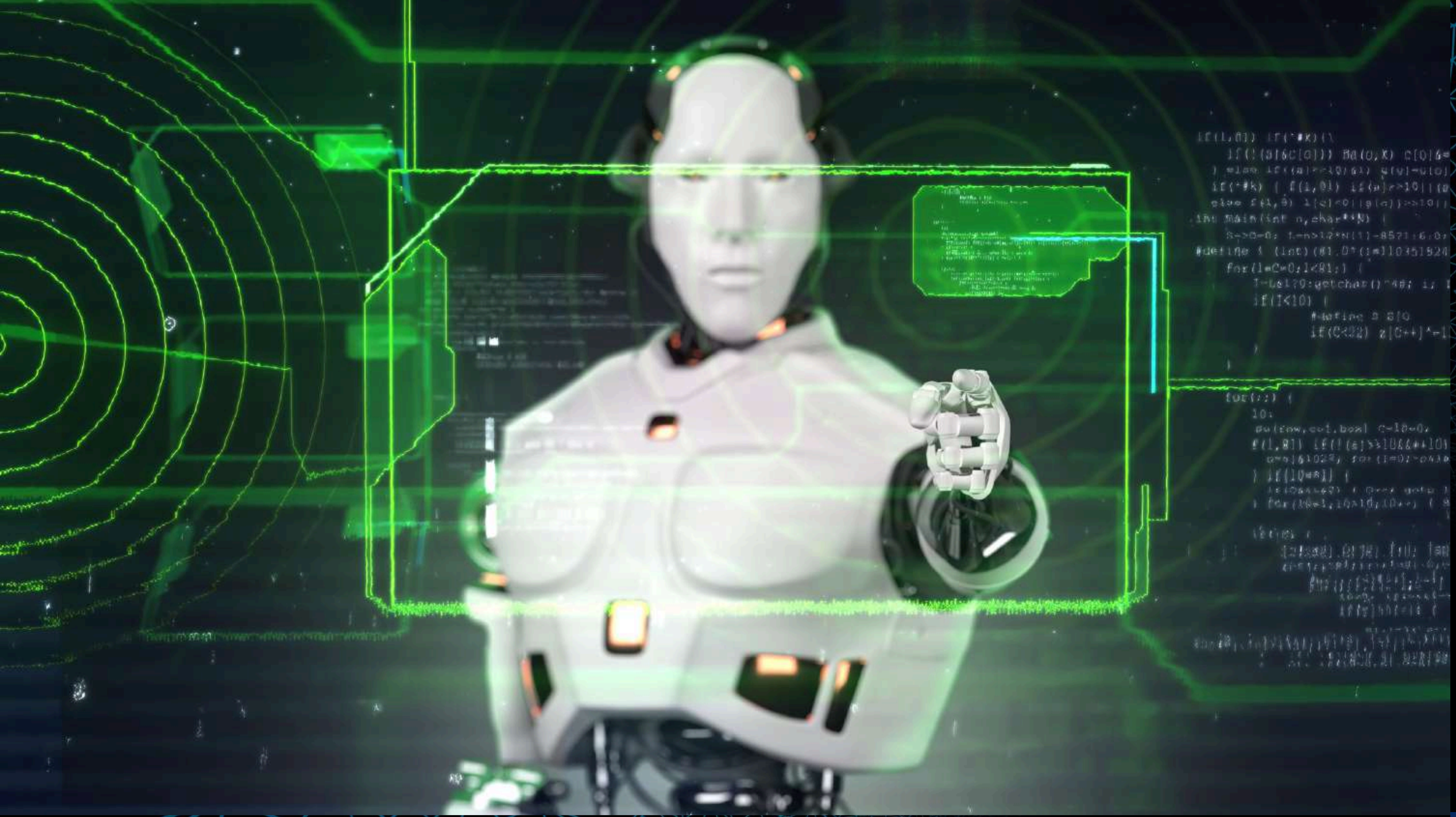
PROBLEM STATEMENT

Fraudulent Account/Transaction Detection

The goal is to identify fraudulent behavior at the transaction or account level using patterns in financial data. The challenge lies in the adversarial nature of fraud, where malicious actors often mimic normal behavior to evade detection. Additionally, the class imbalance—with few fraudulent cases—makes model training difficult.

Last Known Location Inference of Defaulters

The goal is to estimate the last known location of loan defaulters using anonymized, timestamped mobile tower connection logs. The challenge is to model users’ movements over time while preserving temporal consistency and handling data noise and missing values, relying solely on the provided dataset for fraud detection but synthetic data for extracting location of the defaulter.



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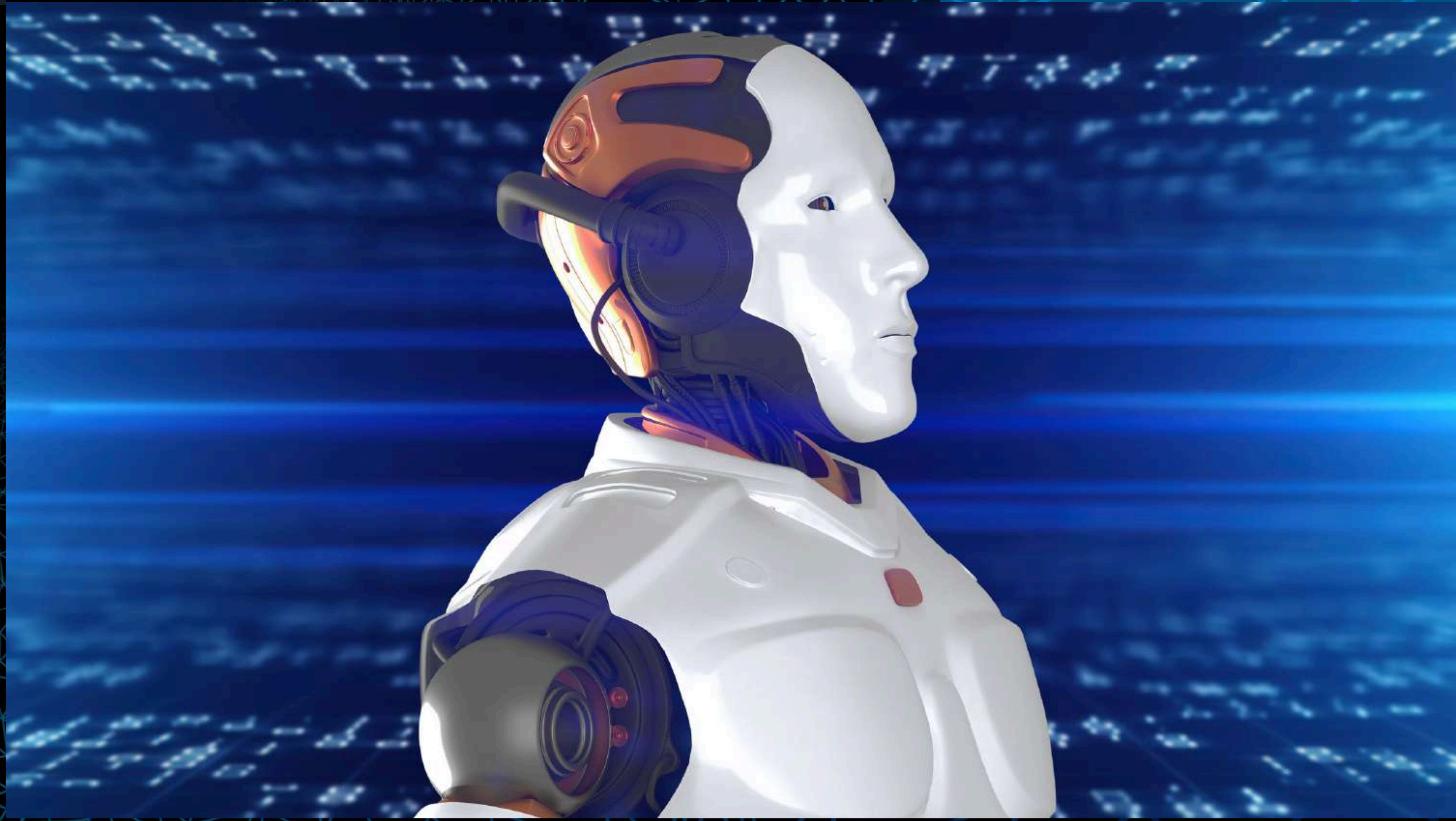
TASK I

It focuses on predicting loan defaulters by analyzing a user's financial, behavioral, and credit profile data.



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DATASET OVERVIEW

Customer Demographics & Identity

Captures basic customer information such as age, income band, and identity verification/compliance indicators (KYC, UID, eKYC).

Credit Bureau Profile

Details the number, status, and performance of credit accounts reported to credit bureaus, including overdue and newly opened accounts.

Loan Portfolio Overview

Summarizes the customer's total loans, sanctioned amounts, outstanding balances, and the age and status of their oldest and newest loans.

Loan & Account Characteristics

Includes details about loan amounts, tenures, EMIs, and account-specific parameters like standing instructions and credit limits.

Delinquency & Risk History

Tracks customer defaults, risk grades (e.g., RG1–RG4), IRAC classification changes, and NPA-related behavior over time.

Behavioral Transaction Patterns

Provides 12-month time series data on credit, debit, average balances, and outstanding trends, reflecting the customer's financial behavior.

Loan Default Indicator

The target variable flags whether a customer defaulted on their loan, forming the basis for supervised learning models.

Credit Score & Inquiry Dynamics

Includes credit scores, credit score changes, and the frequency of credit report inquiries, indicating credit health and activity.

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FEATURE CREATION

Overspend_Ratio

Ratio of total overspend amount to total spending amount across all months

max_consec_overspend

Maximum number of consecutive months where spending exceeded the monthly limit

outbal_slope

Linear regression slope of outstanding balance trend over 12 months

outbal_is_declining

Binary indicator (1/0) showing whether outstanding balance is decreasing

slope_MTD

Linear regression slope of average term debit trend over 12 months

is_debit_declining_MTD

Binary indicator (1/0) showing whether term debit amounts are decreasing

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PROBLEMS FACED

Missing Values

- The dataset has extensive missing values, particularly in credit history and behavioural flag features.
- This complicates analysis, reduces data quality, and may bias model results.

Categorical Columns

- The challenge lies in appropriately encoding these categories.
- For example, preserving order in INCOME_BAND1 while ensuring the nominal variables are numerically represented for modelling, despite their lack of inherent order.

High Dimensionality

- The dataset contained a total of 139 columns, spanning account-level, behavioral, transactional, and derived features.
- The large number of features presented risks such as redundancy, multicollinearity, and overfitting.

Mixed Data Types & Non-Numeric Columns

- Inconsistent binary flags (eg. SI_FLG) contained ambiguous values like 'YN', 'NY', and 'YY' alongside standard 'Y'/'N', risking misclassification during numeric conversion.
- Unstructured duration strings (e.g., "2yrs 3mon") lacked standardized formatting, complicating automated parsing into numeric months without data loss or errors.

Handling Temporal and aggregated features

- Dataset includes monthly time-series features like credit (ONEMNTHCR–TWELVEMNTHCR) debit (ONEMNTHSDR–TWELVEMNTHSDR), and outstanding balances (ONEMNTHOUTSTANGBAL–TWELVEMNTHOUTSTANGBAL)
- Challenges: high dimensionality, multicollinearity, and missing values.

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DETECT FRAUDULENT ACTIVITY

Data Cleaning

- Prepare clean, consistent, and complete data for modeling by handling missing values, transforming distributions, and scaling features.

- Split Data: Used train_test_split with stratification to preserve class balance.
- Power Transformation: Applied Yeo-Johnson transformation to reduce skewness and normalize numeric features.
- Scaling: Normalized features to [0, 1] range using MinMaxScaler.

Feature Optimization

- Balance the dataset, reduce dimensionality, and retain only the most informative features.
- SMOTE-Tomek Resampling: Handled class imbalance by oversampling the minority class and cleaning overlaps using SMOTETomek on the training data.
- Model-based Feature Selection:
- Trained an initial XGBClassifier on the resampled data.
- Used SHAP values to compute average feature importance.
- Selected top 30 most impactful features for downstream modeling.

Model Training

- Train and tune a high-performing model using selected features and cross-validation.
- Hyperparameter Tuning with GridSearchCV and tuned XGBoost using F1 score and 3-fold cross-validation. Search included key parameters: max_depth, learning_rate, n_estimators, subsample, scale_pos_weight.

Post-Processing & Interpretability

- Evaluate performance, derive insights, and prepare for model interpretation or deployment.
- Generated predictions on the test set.
- Evaluated accuracy, confusion matrix, and classification metrics.
- Reported top 20 SHAP features contributing to model decisions.

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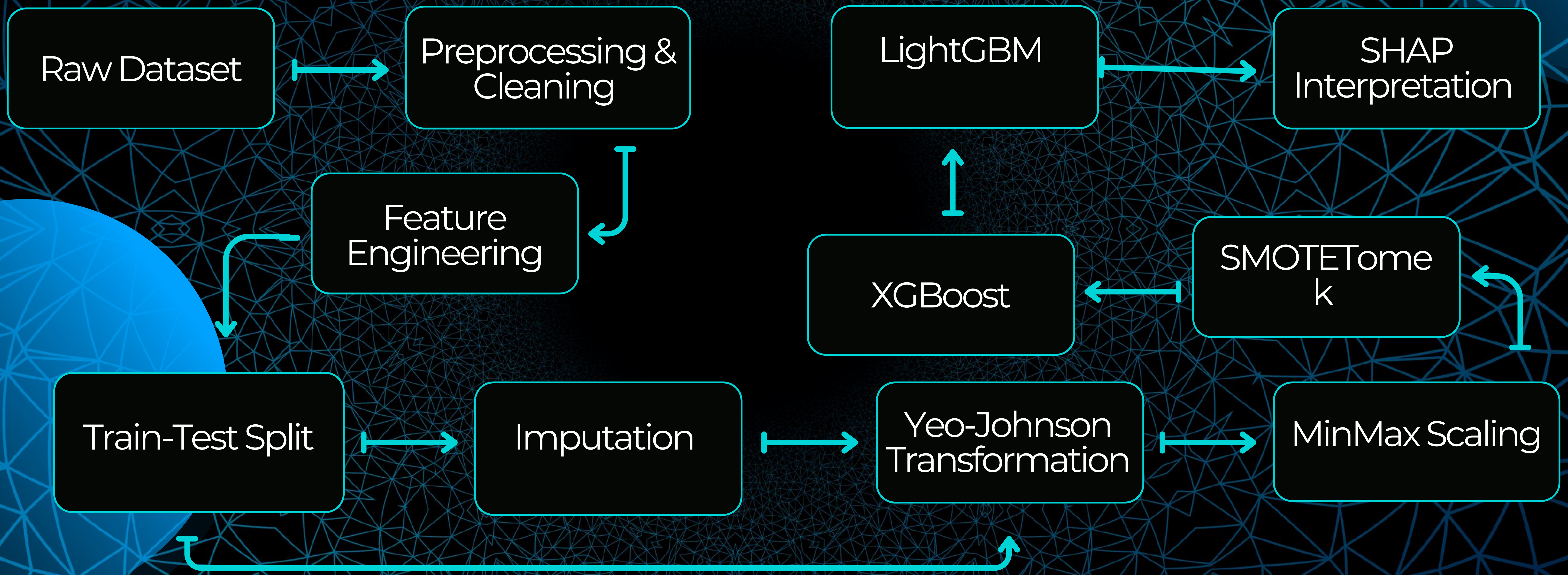
SOLUTION ARCHITECTURE



Data Preprocessing: Imputation, Transformation & Scaling

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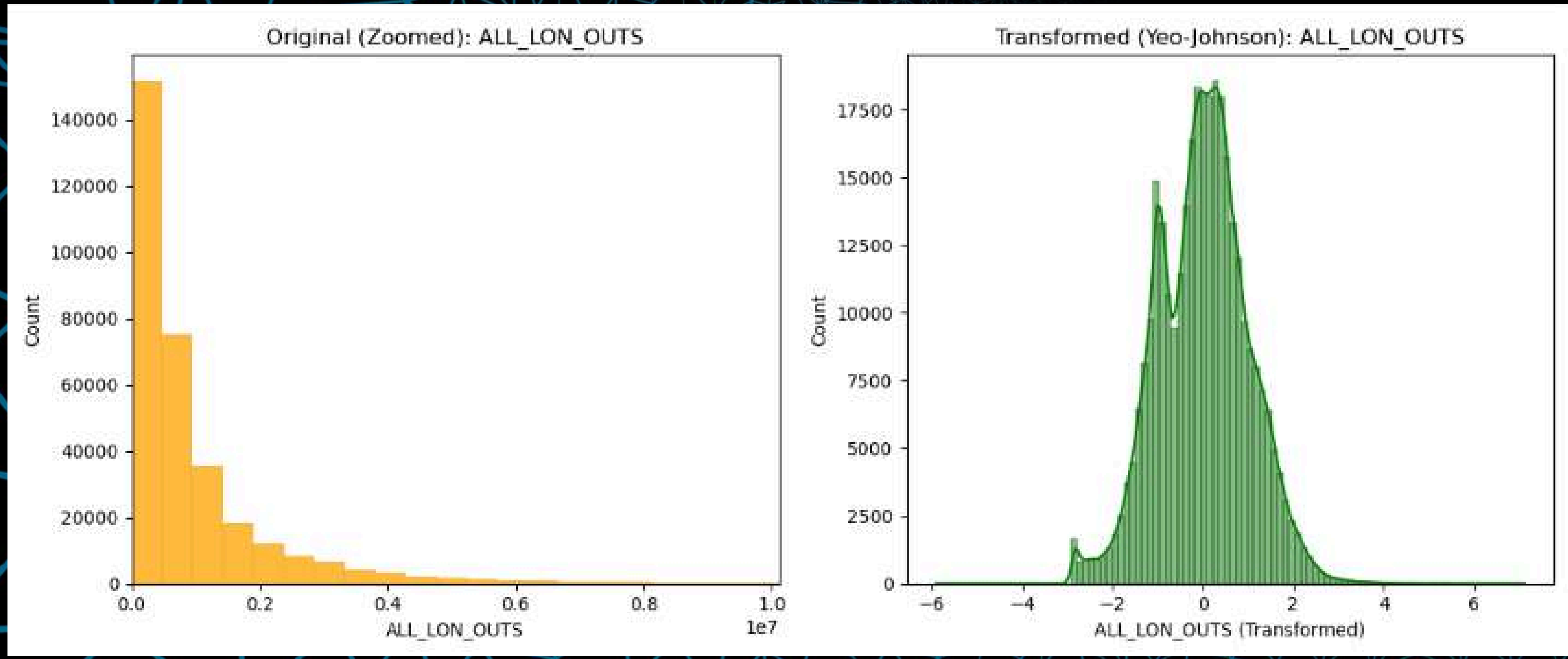
1. Skewed Distribution (Original)
    - Shows that ALL_LON_OUTS is highly right-skewed, indicating extreme loan outliers and non-normality, which can harm model performance.
2. Transformation Applied
    - Used Yeo-Johnson transformation to reduce skewness and stabilize variance without requiring positive-only values (unlike Box-Cox).
3. Transformed Distribution
    - Post-transformation, the feature becomes approximately normal, improving suitability for models sensitive to distribution (e.g., logistic regression, XGBoost).

The original feature exhibited high positive skewness, which was normalized using Yeo-Johnson transformation to improve model interpretability and performance.



To ensure data quality, consistency, and model stability, a structured preprocessing pipeline was implemented. It included handling missing values, transforming feature distributions, and scaling data across a consistent range. These steps were strictly performed after the train-test split to prevent data leakage.

Yeo-Johnson Transformation Formula & Code

Formula:

For a value x and transformation parameter λ:

- If $x \geq 0$:

$$Y(\lambda) = \begin{cases} \frac{(x+1)^{\lambda}-1}{\lambda}, & \text{if } \lambda \neq 0 \\ \log(x+1), & \text{if } \lambda = 0 \end{cases}$$

- If $x < 0$:

$$Y(\lambda) = \begin{cases} -\frac{(-x+1)^{2-\lambda}-1}{2-\lambda}, & \text{if } \lambda \neq 2 \\ -\log(-x+1), & \text{if } \lambda = 2 \end{cases}$$

python

```python
PowerTransformer(method='yeo-johnson', standardize=False).fit_transform(df)
```

Feature Scaling & Class Imbalance Handling

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Feature Scaling (Min-Max Scaler)

Objective:
To normalize all features to a uniform scale (0 to 1), ensuring that no variable dominates the model due to higher magnitude.

Why It Matters:
Algorithms like KNN, logistic regression, and gradient-based methods can perform poorly when features have varying scales.

Method Applied:
Used MinMaxScaler from sklearn.preprocessing on the training set and transformed both train and test sets accordingly.

```python
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

SMOTE-TOMEK Method

[Imbalanced Data]
The original dataset had a skewed class distribution, making it hard for models to learn minority class patterns.

[SMOTE-Tomek] (Add synthetic minority)
SMOTE generates new synthetic samples for the minority class by interpolating between existing instances.

[Balanced Clean Data]
The result is a well-distributed dataset with clearer class boundaries, improving model fairness and performance.

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SHAP-Based Feature Selection with XGBoost

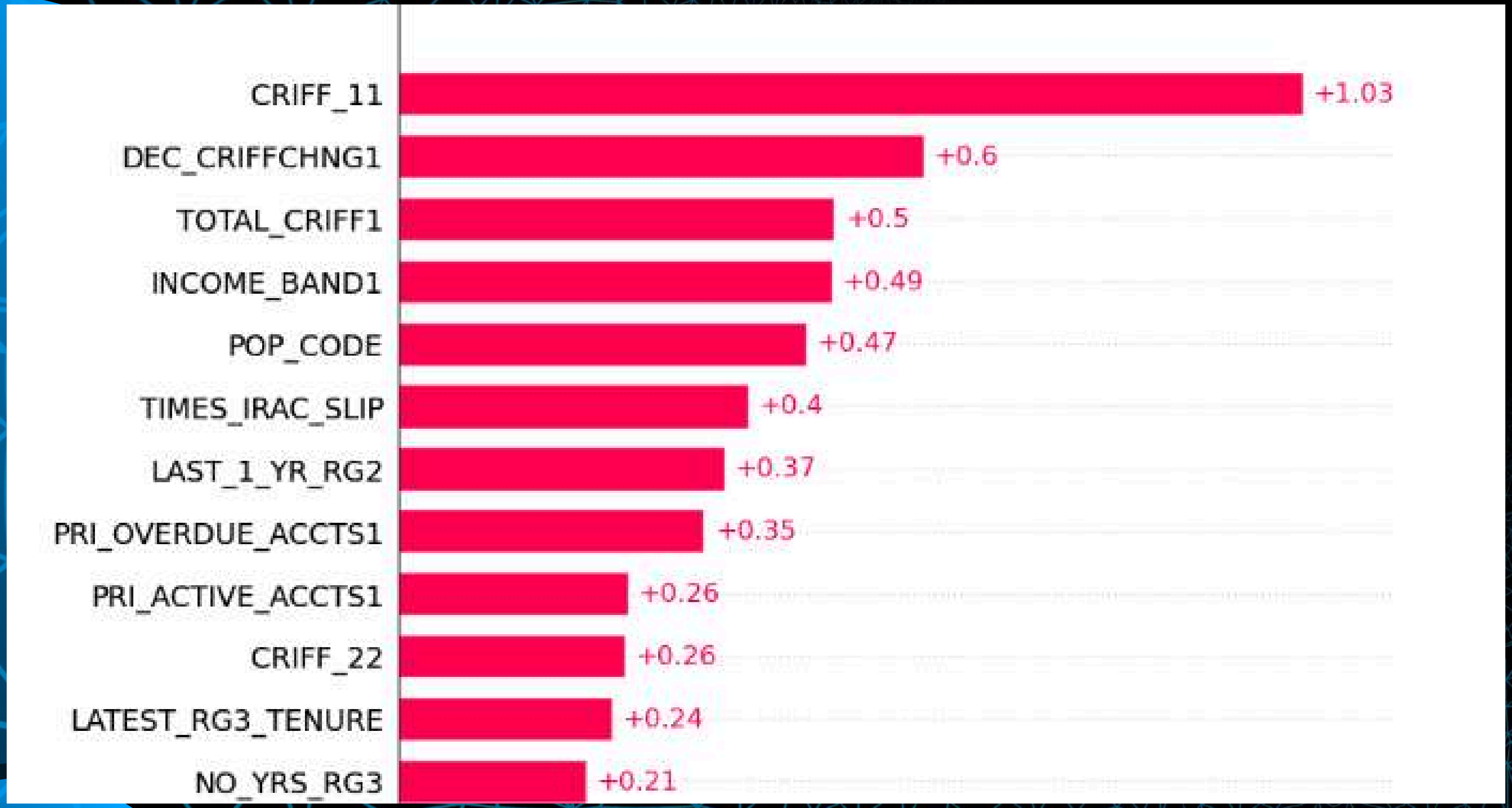
We identify the most influential features using SHAP values derived from an XGBoost model, and use only the top 30 features for final training and evaluation.

Formula of SHAP Value

For a feature i, its SHAP value ϕ_i is:

$$\phi_i(f, x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S) \right]$$

SHAP Values of few features



XGBoost Working

XGBoost offers fast training and works well with SHAP.

Provides model-agnostic feature explanations using TreeExplainer

SHAP values aggregate feature impact consistently across all training samples. Helps identify the top predictive features in a principled way.

MODEL ARCHITECTURE

Feature Selection

Used SHAP values from an XGBoost model to rank features by importance. The top 30 features (e.g., CRIFF_33, OUTS, ACCT_AGE, etc.) were selected to train a more focused and interpretable model.

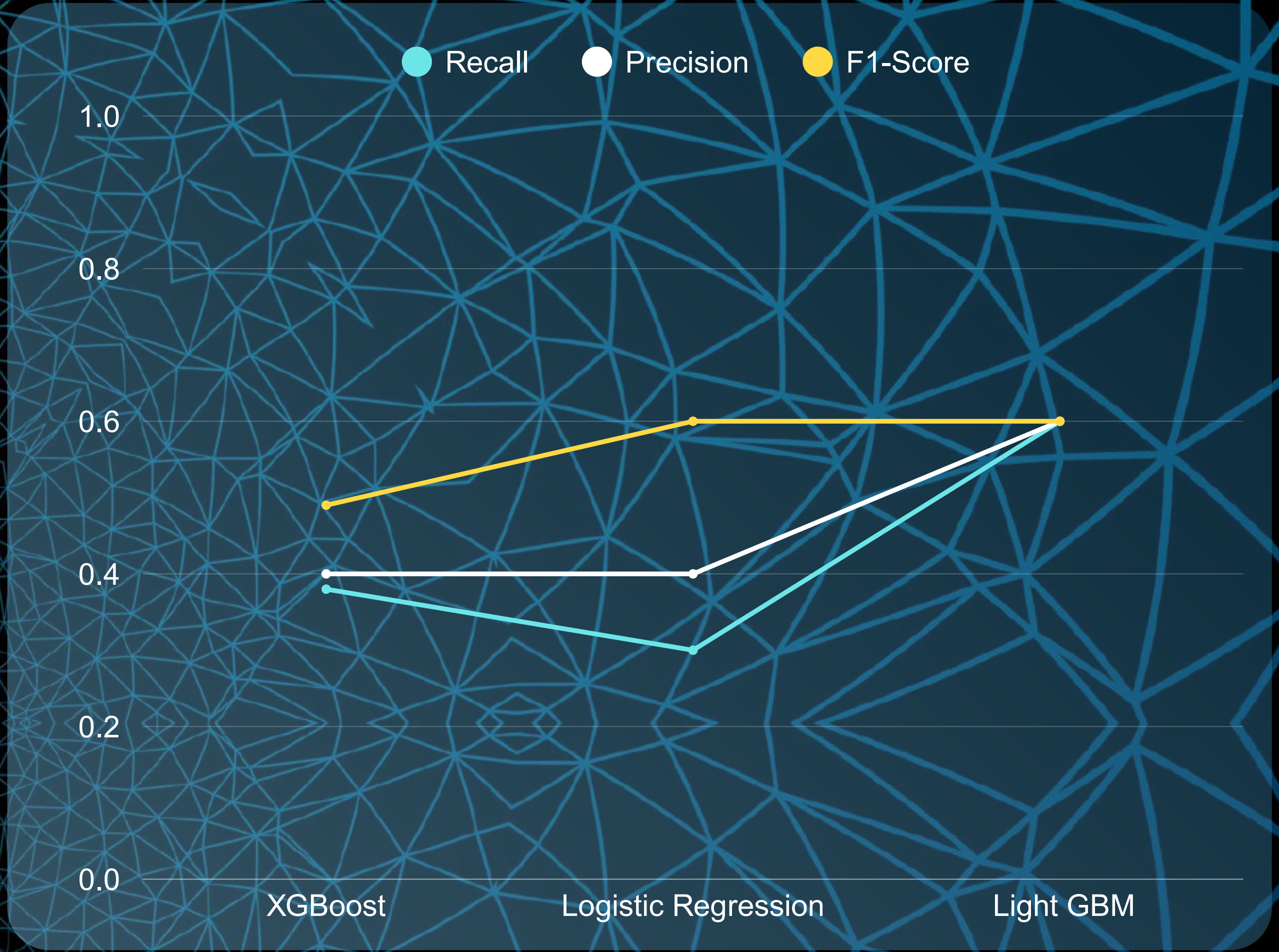
Threshold Optimization

- Predicted class probabilities on the test set and calculated the F1 score
- Optimal threshold ≈ {best_thresh:.2f}
- Achieved F1 Score ≈ {max(f1_scores):.4f},
- improving overall model balance between precision and recall.

Tuned LightGBM with Class Balancing

- Trained a LightGBMClassifier with key parameters:
- num_leaves=31, learning_rate=0.1, n_estimators=500
- subsample=0.8, colsample_bytree=0.8
- reg_alpha=1.0, reg_lambda=1.0, min_child_weight=10
- class_weight='balanced' to handle target imbalance effectively.

Model Performance Comparison



Interpretability

- SHAP Summary Plot ranks top drivers:
  - Credit_utilization
  - trend_slope
  - cancel_count
- Global + Local interpretability
- Stakeholder and compliance ready
- Aligns with Explainable AI (XAI) norms

Precision 0.60

Recall 0.61

F1-Score 0.60

Accuracy 91%

Final Achievement: 91% Accuracy, 61% Recall, 60% Precision with Full Interpretability.

F1 Score: 60%

Future Prospects for Loan Default Prediction

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LLM-based Learning from CSVs

Utilize Large Language Models (LLMs) to read and reason over CSVs directly, allowing the model to learn latent predictors of loan default from raw tabular data in natural language form.

Hybrid Human-AI Model Exploration

Combine rule-based credit risk systems with LLM-generated insights to create interpretable, auditable models that align with regulatory needs.

Natural Language Explainability Layer

Extend the model to generate plain-English rationales (via LLMs) for why a borrower might default, helping credit officers understand and trust model predictions.

Geolocation-Enhanced Risk Modeling

Integrate geospatial behavior and regional economic trends (e.g., urban/rural tags, inflation zones) to enhance model context.

Auto-Feature Generation

Allow LLMs to propose new feature ideas from raw data, accelerating exploratory data analysis and discovery of hidden predictors.

Working Website

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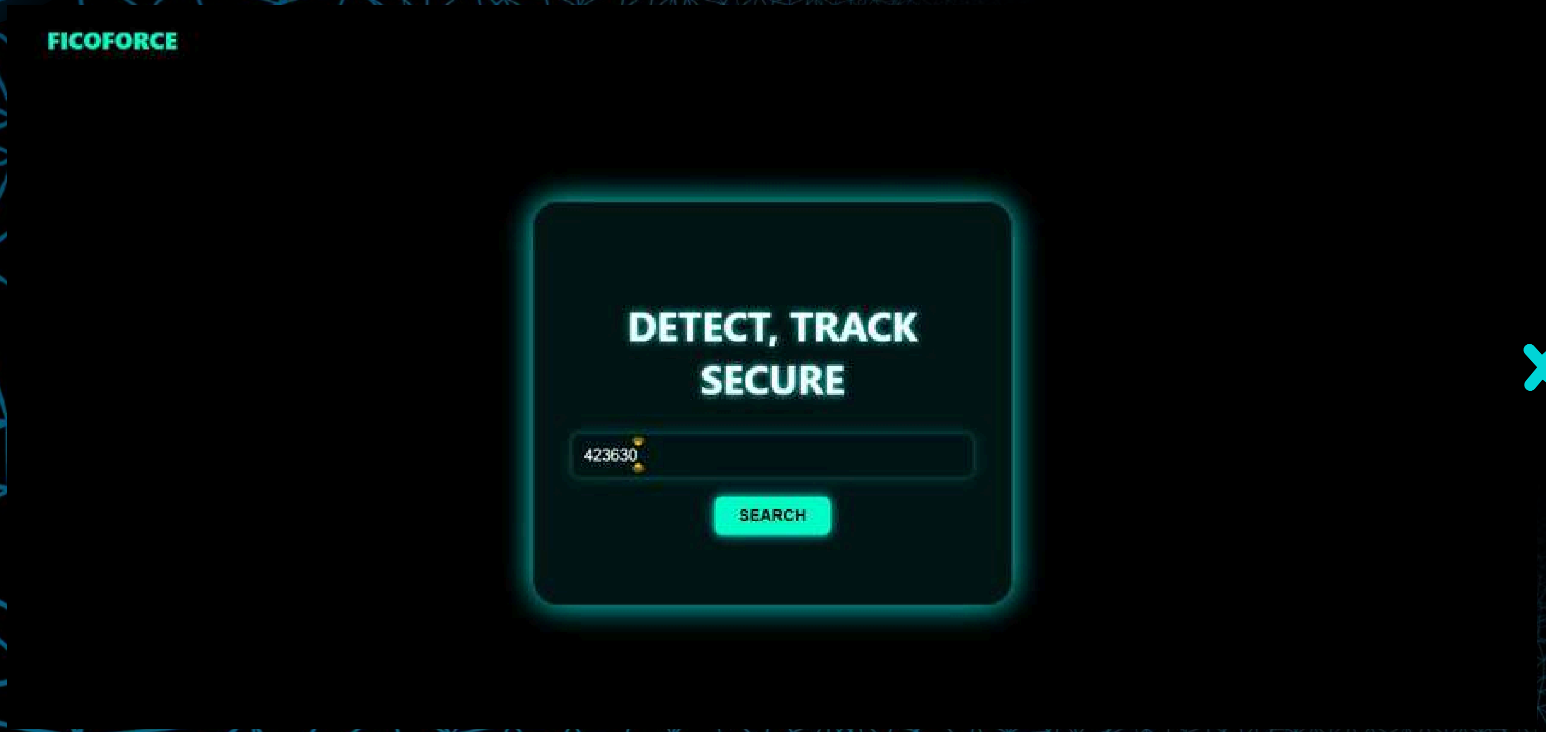

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TASK 2

It focuses on predicting the last known location of loan defaulters using the available user details from the dataset using AI agents.

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MULTI-AGENT LOCATION ANALYSIS SYSTEM

LOCATION PREDICTION WITHOUT GPS

Our task was to estimate the last known location of loan defaulters using only structured fields such as IFSC codes, DL numbers, vehicle registrations, and addresses.
No mobile tower or GPS data was available—making this a pure data inference challenge.

FULLY AUTOMATED TRACKING

To avoid manual effort, we designed an AI-driven system that reads a user dataset (CSV), scrapes platforms like LinkedIn, and generates a per-user location prediction.
The goal was to create a solution that is fast, hands-free, and deployable at scale.

RELIABLE, MULTI-SOURCE VALIDATION

Our multi-agent architecture analyzes signals from static data (documents), behavioral patterns (ATM/UPI), and social sources (LinkedIn).
A conflict-aware scoring engine then validates, ranks, and outputs the most likely location with a confidence score.

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WORKFLOW PROCESS

Static Data

Extracts location clues from account IFSC, DL numbers, vehicle registration, and address fields.

Activity Data

Analyzes behavioral patterns using ATM and UPI transaction locations.

Social Data

Scrapes LinkedIn to identify user-stated current cities or work locations.

F — Data Collection

LinkedIn scraping & dataset preparation

L — Multi-agent Analysis

Specialized agents process different data types

O — Cross Validation

Conflict detection & score calculation

W — Final Output

Location prediction with confidence level

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DATASET OVERVIEW

Social Data

Scraped from public platforms like LinkedIn. Supports the Activity Verifier Agent.

Static Data

Fields tied to official documents or account details. Used by the Static Verifier Agent.

Activity Data

Indicate user's recent interactions or location behavior. Used by the Activity Verifier.

LinkedIn Location

Shows your current city or place of work.

WEIGHT: +1.5

Account Branch

Last six characters of IFSC code indicating bank branch location

WEIGHT: +2

DL Number

Driving license number with state identification code

WEIGHT: +2

Vehicle Number

Vehicle registration number with state code

WEIGHT: +1.5

Address

Residential address with city and state information

WEIGHT: +1.5

ATM Transactions

Location of recent ATM withdrawal transactions

WEIGHT: +3

UPI Location

Location of recent UPI payment transactions

WEIGHT: +2.5

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DATASET OVERVIEW

Activity Data

Indicate user's recent interactions or location behavior. Used by the Activity Verifier.

Frequent Location

Shows Frequent Location

WEIGHT: +1

Last Location

Shows Last Location of that person based on mobile app

WEIGHT: +1

Rules

- High: score ≥ 7
- Medium: 4 ≤ score < 7
- Low: score < 4
- Low or 3+ cities or critical missing → append "(Manual Review Needed)"
- +0.5 bonus if home and last known in same

Apply -2 penalty per state conflict

- Note if DL + Branch both missing
- Note if 3+ distinct cities exist
- Check if home and last known are in same state
- Give raw final score for home location after penalty

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WHY AI AGENTS?

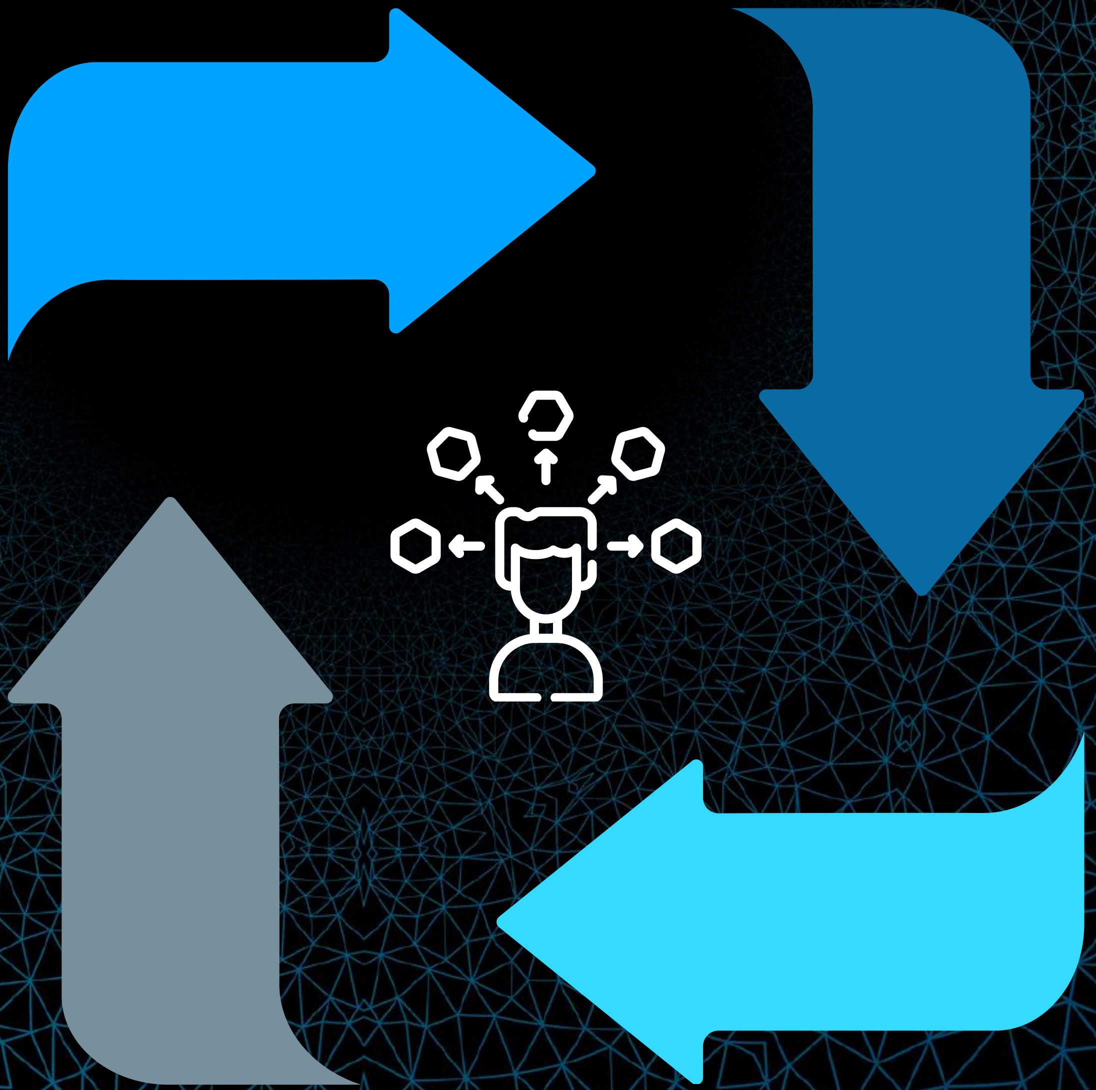
Instead of one-size-fits-all logic, we designed four specialized AI agents, each focused on a specific aspect of location analysis. This ensures better accuracy, explainability, and conflict handling.

Static Verifier

Analyzes fixed fields like IFSC code, DL number, vehicle number, and address to derive probable location.

Activity Verifier

Checks recent location-related activity from ATM/UPI transactions and social sources like LinkedIn.



Cross Validator

Compares outputs from both agents, detects conflicting locations, and applies penalties or bonuses accordingly.

Scoring Evaluator

Combines all insights to predict the final location and assigns a confidence level: High / Medium / Low.

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AGENTS WORKFLOW

AGENT 1: STATIC VERIFIER

Identify the user's likely location using permanent, document-based data.
This agent focuses on structured fields that carry embedded geographic information.

- IFSC Code (Bank Branch)
- Driving License Number (State Prefix)
- Vehicle Registration Number (State Code)
- Residential Address (City/State Extraction)
- Phone Number Prefix (Regional Hint)

Why It Matters:
These fields provide official and registered locations, serving as a stable baseline for inference.

AGENT 2: ACTIVITY VERIFIER

Capture the user's recent and dynamic location patterns through behavior and digital presence.

- ATM Transaction Locations
- UPI Payment Locations
- LinkedIn Location (Scraped)
- Recent/Frequent Location Fields

Why It Matters:
These indicators reflect the user’s actual movement and current presence, helping update or validate the static predictions.

AGENTS WORKFLOW

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AGENT 3: CROSS VALIDATOR

Compare predictions from the Static and Activity agents, detect inconsistencies, and resolve conflicts to ensure the final location is both reliable and explainable.

Core Responsibilities:

- Identifies location mismatches and applies conflict penalties
- Adds bonuses for consistency (e.g., same state across sources)
- Flags incomplete or ambiguous cases for manual review

Why It Matters:
Ensures the final prediction is logically sound and not based on isolated or conflicting signals.

AGENT 4: SCORING EVALUATOR

Calculate a final score for each user's predicted location and assign a confidence level (High / Medium / Low) based on combined weighted signals and penalties.

Scoring Rules:

- High Confidence: Score ≥ 7
- Medium Confidence: 4–6.9
- Low Confidence: < 4
- Triggers manual review if location sources are unclear or conflicting

Output Format:

- **Final Predicted Location**
- **Confidence Level**
- **Last Known Location (if different)**

Why It Matters:
Delivers a clear, interpretable output with confidence scoring that supports decision-making at scale.

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OUTPUT FORMAT & RESULT INTERPRETATION

Each report summarizes all detected locations, their respective confidence scores, and clearly states the system's final prediction.

Sample Output

```
=================== [7/35] Dhanuk Taran ===================

--- Summary Report ---
Person: Dhanuk Taran

Detected Locations:
  Gaya, Bihar - 4.5
  Ramagundam, Telangana - 0
  Kolkata, West Bengal - 0
  Amaravati, Andhra Pradesh - 0
  Gaya, Bihar - 3
  Ramagundam, Telangana - 1
  Kolkata, West Bengal - 1
  Amaravati, Andhra Pradesh - 2.5

Chosen Home: Gaya, Bihar
Chosen Last Known: Amaravati, Andhra Pradesh
Final Prediction: Gaya, Bihar
Confidence: Low (Manual Review Needed)
--------------------------
```

What the Output Shows

- Location Candidates: All possible locations detected across sources with their weighted scores
- Chosen Home & Last Known: Extracted from user data (address, last active field, etc.)
- Final Prediction: Based on aggregate score, adjusted by penalties or bonuses

Confidence Score

- High: score ≥ 7
- - Medium: 4 ≤ score < 7
- - Low: score < 4
- - Low or 3+ cities or critical missing → append "(Manual Review Needed)"

- +0.5 bonus if home and last known in same state

Why This Format Works:

Human-readable: Clearly structured and easy to interpret

Traceable: Shows exactly how the final prediction was derived

Review-friendly: Low-confidence cases are flagged early

Batch-ready: Can be generated for hundreds of users with uniform structure

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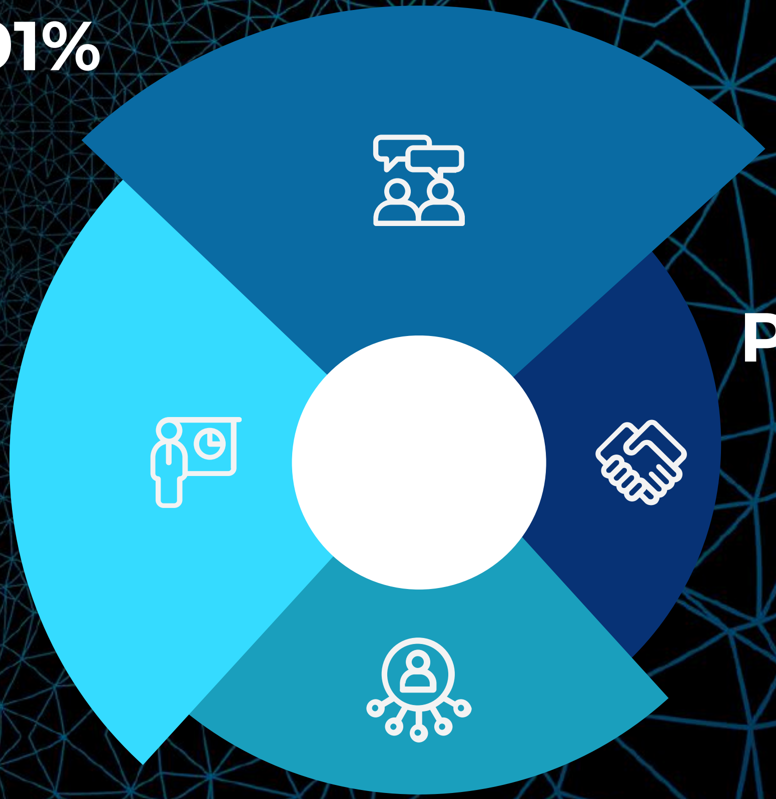
CONCLUSION

Our end-to-end solution addresses both **loan default prediction** and **defaulter location inference** by leveraging structured, behavioral, and social data through interpretable machine learning and multi-agent intelligence—delivering high-accuracy results with full scalability, automation, and decision transparency.

Final Result

Our model **is a combination of XGBoost, Yeo-Johnson transformation, Min-Max Scaling** and LightGBM model and achieved:

Accuracy: 91%



Precision: 60%

Recall: 61%

F1 Score: 60%

THANK-YOU

Team Members

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Avani	Mansi
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